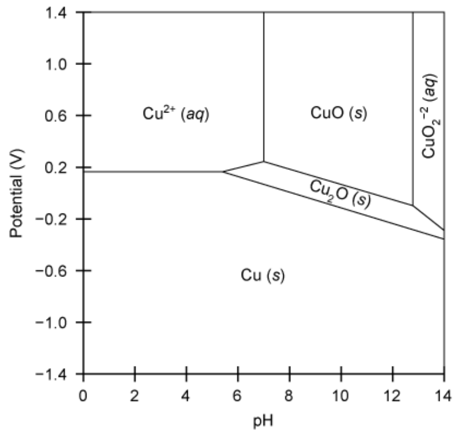
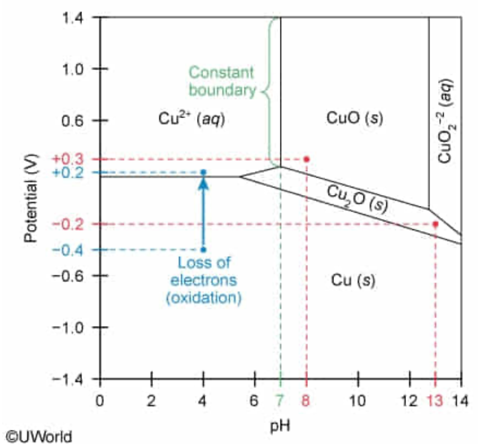
Copper plays a vital role as a trace mineral for biological processes, including the electron transport chain and collagen synthesis. The Pourbaix diagram below outlines the relationship between several copper species as a function of solution pH and applied potential. Which statement accurately describes the chemical changes shown in this diagram?



- A. The predominant species at a potential of −0.2 V and pH of 13, is CuO. [3%]
- B. Cu^{2+} is reduced and precipitates as Cu_2O as pH increases to a pH of 8 at a potential of +0.3 V. [12%]
- C. Cu is reduced to Cu^{2+} as the potential is changed from −0.4 to +0.2 at a pH of 4. [18%]
- ✔ D. The equilibrium between Cu^{2+} and CuO is independent of applied potential above +0.3 V at a pH of 7. [66%]

Correct
66% answered correctly

Explanation:



In the **Pourbaix diagram**, the **boundary lines** between the **species domains** indicate the **transition** from one species into another. Along the boundary between an aqueous and a precipitating solid species, equilibrium can exist.

The domain boundary between Cu^{2+} and CuO (at pH of 7) is a vertical line parallel to the *y*-axis (potential). Above +0.3 V, this line is a constant and does not change regardless of increasing potential. Therefore, the equilibrium between Cu^{2+} and CuO is **independent** (unchanging) with respect to applied potentials above +0.3 V.

(Choice A) The intersection at −0.2 V and pH of 13 falls within the domain of Cu_2O rather than that of

CuO. Therefore, Cu_2O is the predominant species of copper under these conditions.

(Choice B) At a potential of +0.3 V and pH of 8, Cu^{2+} precipitates as CuO rather than Cu_2O. In this process, the oxidation state of Cu^{2+} is unchanged and Cu^{2+} is not reduced.

(Choice C) At pH of 4, moving from an applied potential of −0.4 V to +0.2 V causes Cu to convert to Cu^{2+} (Cu loses two electrons). This is an oxidation rather than a reduction.

Educational objective:
Plots such as Pourbaix diagrams show the relationship between chemical species as a function of given conditions. Boundary lines between domains indicate the transition from one species into another, and can also indicate equilibrium between aqueous and precipitating species. Boundaries parallel to an axis show constant ranges that are independent of changes in the respective axis variable.

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